

SOFTWARE ENGINEERING LAB -1

Problem Statement #30: Airport Lost Luggage Claim & Tracking Portal

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall cross-reference passenger lost baggage reports against unrouted luggage scan logs across all connected airport terminals.	High	Pass: Match identified and recovery alert generated. Fail: Incorrect bag tag format accepted.	Core recovery function; enables automated matching instead of manual agent search.
FR-002	Functional	The system shall allow a Passenger to submit a lost baggage claim by entering bag tag ID, flight number, last-seen location, and a description of the bag.	High	Pass: Claim record is created with a unique Claim ID and status 'Open' within 5 seconds of submission. Fail: Claim submitted without a valid bag tag ID or flight number.	Captures the minimum data needed to begin the recovery search.
FR-003	Functional	The system shall display to the Passenger a real-time timeline of bag tag scan events (check-in, transit hub, gate) for any active claim or checked bag.	Medium	Pass: Timeline shows all scan events in chronological order with terminal and timestamp. Fail: Timeline omits or misorders a recorded scan event.	Gives passengers self-service visibility and reduces support-desk load.
FR-004	Functional	The system shall automatically calculate and initiate a compensation payment when a claim remains unmatched 21 days after the reported date, per airline compensation policy.	High	Pass: Compensation record is generated and routed for payout exactly on day 21 if the claim status is still 'Open'. Fail: No compensation record generated after day 21, or generated before day 21.	Ensures regulatory/contractual compensation deadlines are met without manual tracking.
FR-005	Functional	The system shall allow a Baggage Service Agent to update a claim's status (Open, Matched, Compensated, Closed) and attach investigation notes.	Medium	Pass: Status change is saved, timestamped, and reflected on the Passenger's tracking view within 1 minute. Fail: Status change is not persisted or not visible to the Passenger.	Keeps passenger-facing status in sync with agent-side investigation work.
NFR-001	Nonfunctional	The baggage tracking timeline API must return complete multi-leg scan history in under 200 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures a responsive tracking experience for passengers during high-traffic periods.
NFR-002	Nonfunctional	The system shall encrypt all passenger personally identifiable information (name, passport number, payment/compensation details) at rest using AES-256.	High	Pass: A storage/security audit confirms all PII fields are AES-256 encrypted at rest. Fail: Any PII field found stored in plaintext.	Protects sensitive passenger data and supports data-protection compliance.